

JavaScript Quiz App – Mistakes & Corrections

Printable reference sheet of the code errors you wrote and how to fix them

1. Event Listener – Calling the function immediately

This is the main mistake in the `addEventListeners()` method.

■ WRONG (what you wrote)

```
buttons.forEach(button => {
  button.addEventListener("click", this.checkAnswer(button));
});
```

Why it is wrong:

- `this.checkAnswer(button)` runs the function **immediately** when the page loads.
- The return value of that call (usually `undefined`) is what gets registered as the event listener.
- As a result, clicking the buttons later does nothing useful.

■ CORRECT version

```
buttons.forEach(button => {
  button.addEventListener("click", () => this.checkAnswer(button));
});
```

Why this works:

- The arrow function `() => this.checkAnswer(button)` is passed as the listener.
 - It only runs when the user actually clicks the button.
 - `this` stays correctly bound to the class instance.
-

2. Full corrected `addEventListeners()` method

```
addEventListeners() {
  const buttons = document.querySelectorAll(".answer-btn");

  buttons.forEach(button => {
    button.addEventListener("click", () => this.checkAnswer(button));
  });

  document.addEventListener("keydown", this.handleKeydown);
}
```

3. `handleKeydown()` – This part was already correct

Your keyboard handler was written correctly. It lets the user press keys 1–4 to select answers.

```
handleKeydown(e) {
  const validKeys = ["1", "2", "3", "4"];
  if (validKeys.includes(e.key)) {
    const index = parseInt(e.key) - 1;
    const buttons = document.querySelectorAll(".answer-btn");
    if (buttons[index]) {
      this.checkAnswer(buttons[index]);
    }
  }
}
```

4. Checking the answer (case-insensitive)

This line is also correct and useful:

```
button.dataset.answer.toLowerCase() === this.correctAnswer.toLowerCase()
```

It compares the selected answer with the correct answer while ignoring capital letters (so "Paris", "paris" and "PARIS" are all treated the same).

5. playTone() – Sound feedback (correct)

Your tone function is well written. Use it like this for feedback:

```
// Correct answer – high pitch  
this.playTone(880, 0.2);
```

```
// Wrong answer – low pitch  
this.playTone(200, 0.4);
```

Quick Summary – What to remember

Mistake	Fix
Writing <code>this.checkAnswer(button)</code> inside <code>addEventListener</code>	Wrap it in an arrow function: <code>() => this.checkAnswer(button)</code>
Forgetting that the function runs immediately	Always pass a function reference (or arrow function), never the result of a call

Generated as a study/print sheet from your quiz app code questions.